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Data Science Fundamentals

Decision Trees

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Before we get started…

Practicalities

Learning objectives

- You understand and apply different types of regression, know when and how to use them, can execute them and put them in to practice as well as interpret the results.
- You recognize overfitting in regression models and learn techniques to prevent overfitting.
- You learn about different clustering techniques and understand when and how to use them for grouping data.
- You gain an understanding of dimensionality reduction techniques and learn how to apply them to compress and visualize data.

Course material

- **Canvas**
  - Slides & exercises from classes
- **Online**
  - https://www.w3schools.com/datascience/
- **Software**
  - JupyterLab (or work via Anaconda/VS Code if you prefer)
- **The Internet**



Course breakdown

- **Course Structure:**
  - Lectures: 3 hours per week (theory and exercises)
  - Attendance: Mostly not compulsory, but strongly recommended.

- **Examination Breakdown:**
  - 30%: Continuous assessment (mandatory excercises)
  - 70%: Final exam (oral, mostly theoretical with some practical elements)

Is it difficult to pass?

It’s perfectly doable **if you put in the work**

- Come to class
- Participate in the lessons
- Make your exercises



How to be an ideal student

- **Attend classes** - even when you party hardy
- **Show respect** - towards your instructor and fellow students
- **Be punctual** - or you'll find the door closed
- **Meet deadlines** - late submissions will earn a solid zero
- **Have a question?** - just ask! I don't bite
- **Know it all?** - help your classmates, they might repay you with pizza
- **Do your own work!** - shortcuts will lead to a short academic career ;)

Syllabus

Week - 1	Introduction & Python/JupyterLab Setup	Week - 9	Easter Holiday
Week - 2	Regression - Linear regression	Week - 10	Easter Holiday
Week - 3	Regression - Polynomial regression	Week - 11	Clustering - Hierarchical clustering
Week - 4	Classification - Logistic regression	Week - 12	Dimensionality reduction: Feature Analysis - PCA
Week - 5	Classification - Decision trees	Week - 13	Regularization: Lasso regression
Week - 6	Classification - kNN	Week - 14	Recap: Theory + Excercises
Week - 7	Clustering - kMeans	Week - 15	Exam Preperation + Q&A
Week - 8	Lifesnaps guidance		

(subject to change)

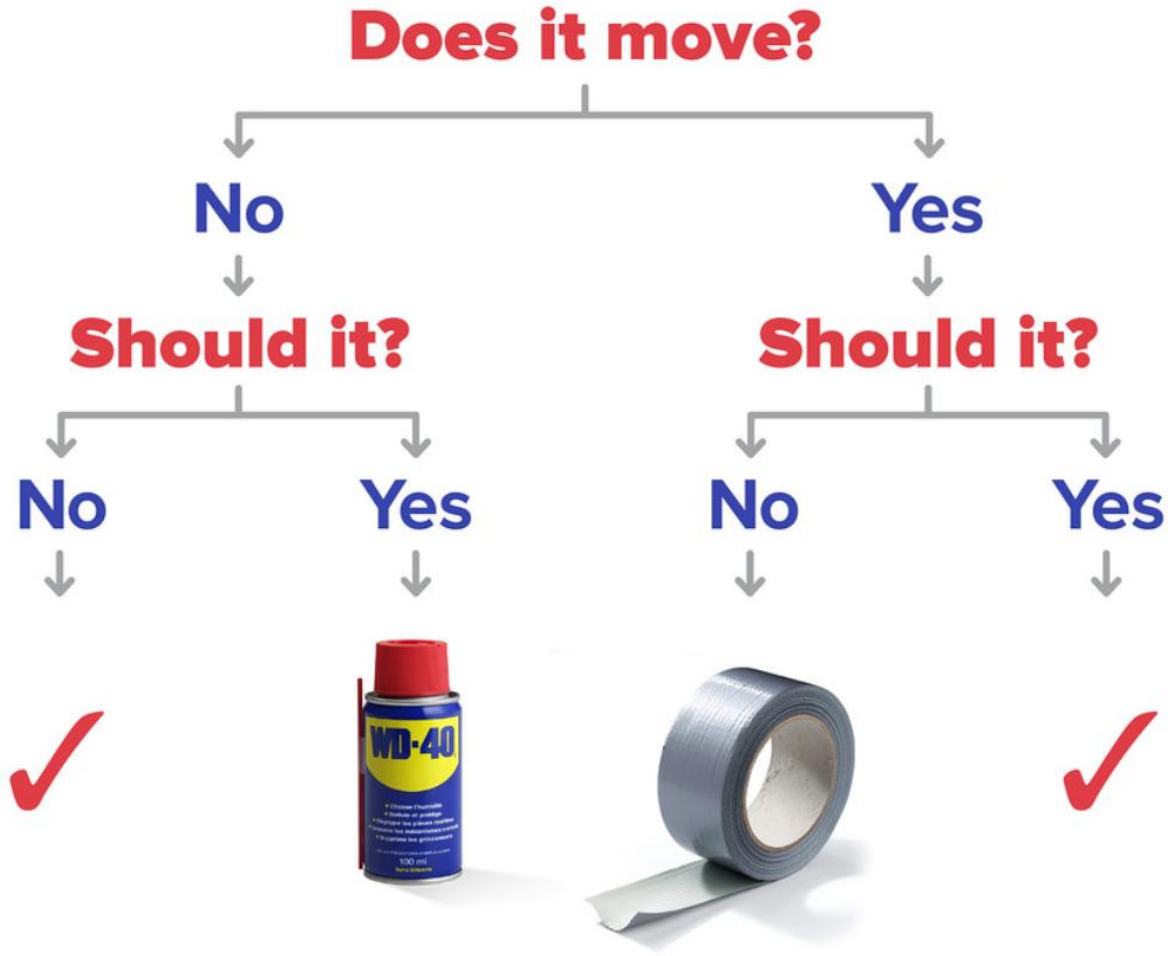
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Now we get to the fun part

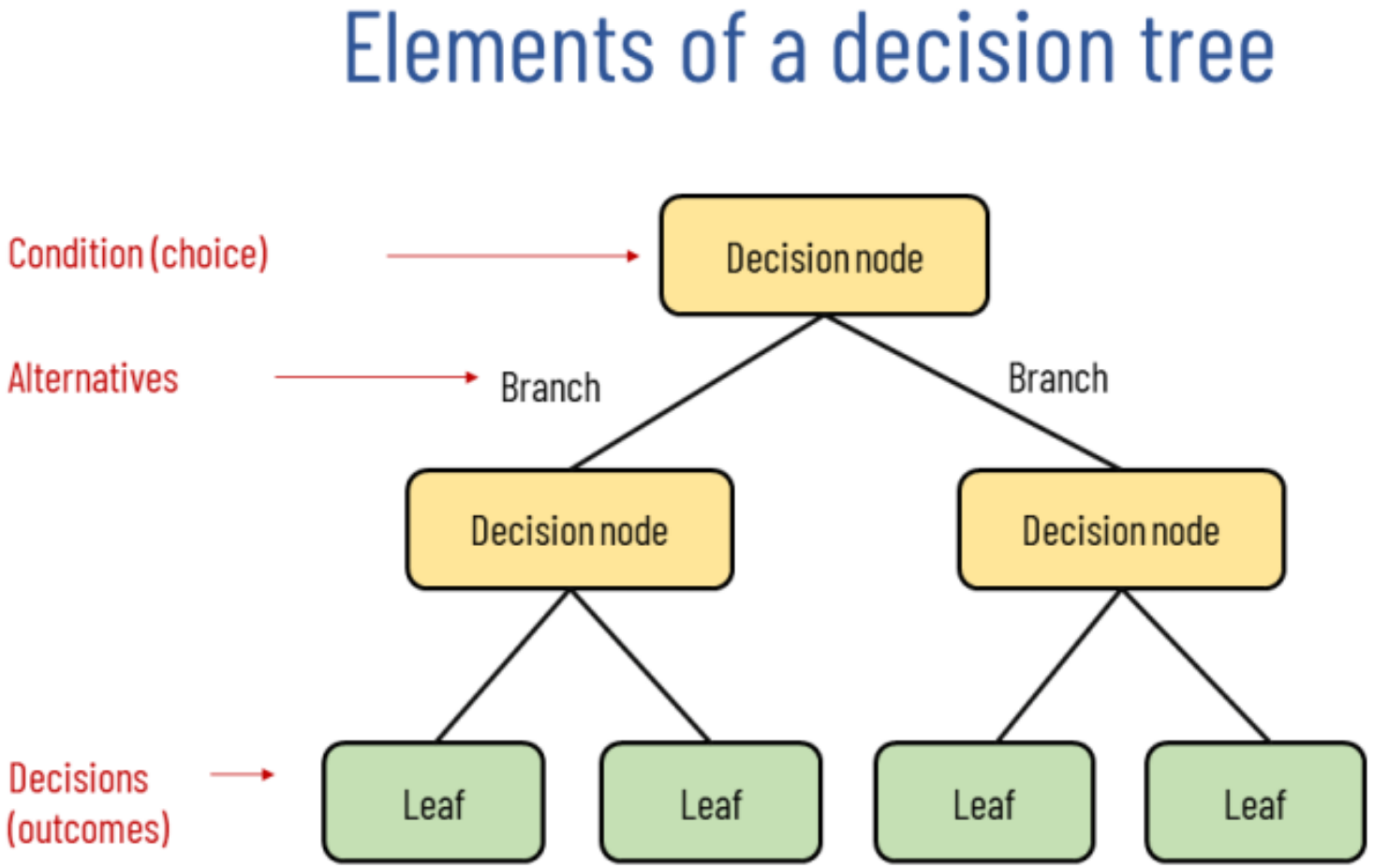
Decision Trees

Decision trees

- Basically a flowchart
- Mimics **human decision** making
- Sever ‘splits’ where you ask a question
- Based on the answer you ask further questions
- Keep going till you end up with your answer!



Terminology



Inner workings

- Each split is based on **one** or **two** variables
- Split so as to maximize information gain
  - Each split **must** deliver information gain
- We keep asking questions, but hold two important things in our mind:
  - How is the split condition determined?
  - When should we stop?
  - → see lecture Regularization

Overfitting

We don’t have to worry about this since it’s classification and python will take care of this right?

Actually, we do…



Recognize overfitting

- Overfitting occurs in **all** data science models, even in classification
- Also in Logistic regression, even though we (mostly) glossed over it
- A model can ask **too many** questions/**the wrong** questions
- Compare **testing** accuracy vs **training** accuracy



High accuracy on training data

+ low accuracy on testing data

= overfitting!

Combat overfitting

- Same fix as ploymial regression:
  - Keep polynomial degree low
  - Keep amount of nodes low
- The **fewer questions** a model asks, the **more** it needs to **learn** from those questions
- Learning trends > recognizing values



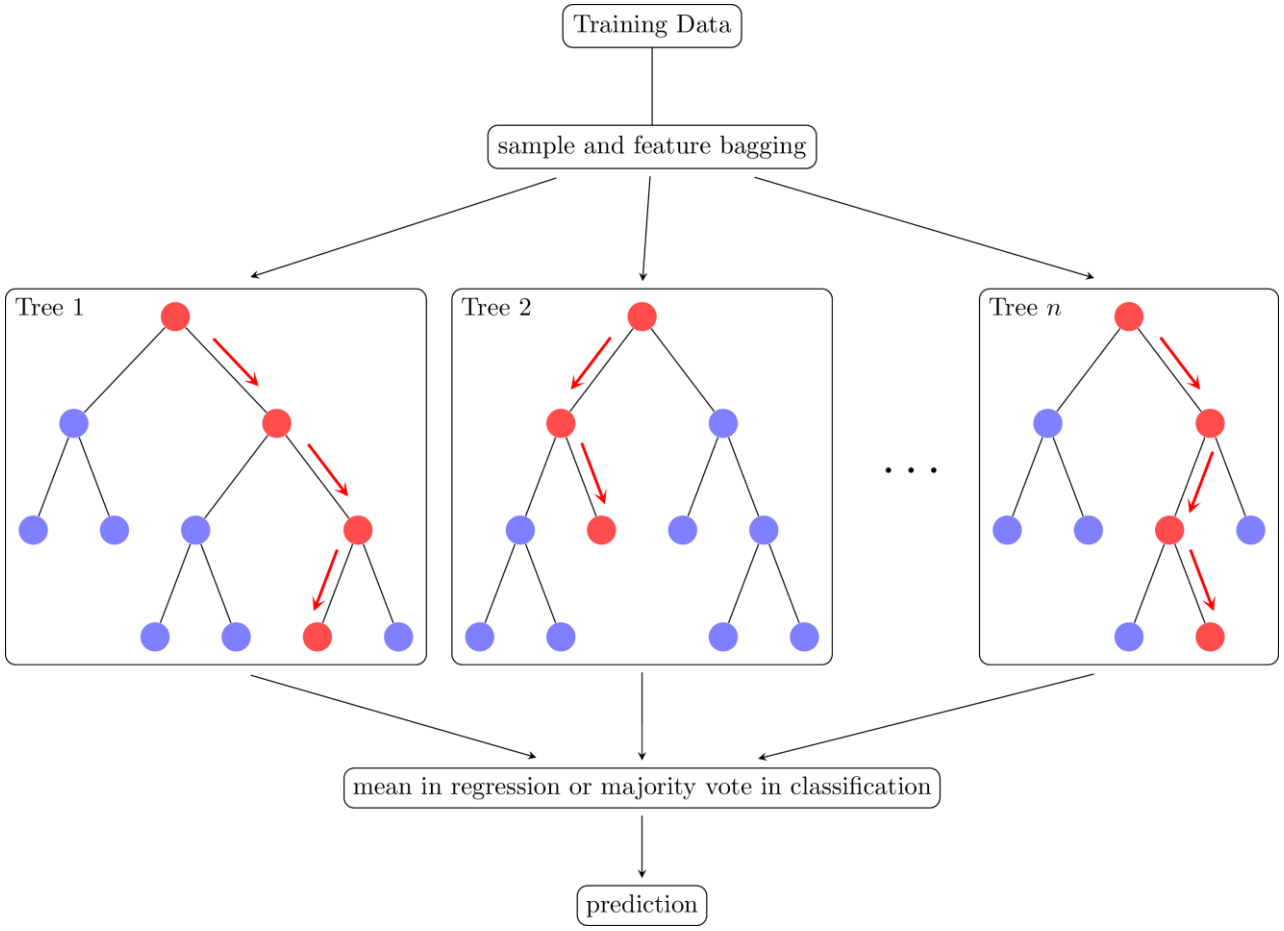
Metrics

- The same classification metrics apply:
  - Accuracy
  - Precision
  - Recal
  - Specificity
  - F1 Score



Random forest

- Extended form of a decision tree
- Chain of decision trees
- Each tree:
  - Gets a **subset** of data
  - Makes a prediction
- Majority vote
- Can also be used for **regression**!



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Let's jump into action

Excercises

First, let's explore this together

- Go to canvas and download the notebook "DSF 05 – Decision Trees Notebook"
- Open the notebook in Jupyter
  - Open a CMD and run "jupyter lab"
  - Or launch it via anaconda
- Follow along!



Now it's up to you!

- Go to canvas and download the notebook "DSF 05 – Decision Trees Exercise"
- Also download the csv file called "Wine.csv"
- Place them in the same directory
- Open the notebook in Jupyter
  - Open a CMD and run "jupyter lab"
  - Or launch it via anaconda
- Go through the notebook and make the excercises

I don't bite, ask me any questions you have!

